Questions: Arithmetic on numerical fractions

Donald Campbell

Summary

A selection of questions for the study guide on arithmetic on numerical fractions.

Before attempting these questions, it is highly recommended that you read Guide: Arithmetic on numerical fractions. You may also find Calculator: Highest common factor, lowest common multiple useful.

Q1

Calculate the following additions and subtractions by first finding a common denominator. Write your answer in its simplest form.

1.1. $\frac{3}{8}+\frac{1}{8}$

1.2. $-\frac{2}{7}+\frac{5}{7}$

1.3. $\frac{13}{20}-\frac{17}{20}$

1.4. $\frac{1}{3}+\frac{1}{6}$

1.5. $\frac{3}{4}-\frac{1}{2}$

1.6. $\frac{2}{5}+\frac{1}{10}$

1.7. $\frac{5}{6}-\frac{1}{3}$

1.8. $\frac{1}{2}+\frac{3}{10}$

1.9. $\frac{7}{10}+\frac{2}{15}$

1.10. $\frac{3}{8}+\frac{5}{10}$

1.11. $\frac{3}{4}-\frac{2}{5}$

1.12. $-\frac{2}{3}+\frac{3}{4}$

1.13. $\frac{9}{10}-\frac{1}{15}$

1.14. $-\frac{7}{12}-\frac{1}{15}$

1.15. $\frac{11}{14}+1$

Q2

Calculate the following multiplications. Write your answer in its simplest form.

2.1. $\frac{1}{4}\cdot\frac{1}{3}$

2.2. $\frac{2}{5}\cdot\frac{3}{7}$

2.3. $\frac{3}{5}\cdot 10$

2.4. $\frac{5}{8}\cdot\frac{4}{15}$

2.5. $\frac{2}{3}\cdot\frac{9}{10}$

2.6. $-\frac{1}{7}\cdot\frac{3}{4}$

2.7. $\frac{6}{11}\cdot\frac{22}{3}$

2.8. $8\cdot\frac{5}{12}$

2.9. $-\frac{7}{9}\cdot\frac{6}{5}$

2.10. $\frac{4}{5}\cdot\left(-\frac{15}{16}\right)$

2.11. $\frac{12}{13}\cdot\frac{1}{6}$

2.12. $\left(-\frac{9}{10}\right)\cdot\left(-\frac{5}{12}\right)$

Q3

Calculate the following divisions. Write your answer in its simplest form.

3.1. $\frac{1}{3}\div\frac{1}{6}$

3.2. $\frac{2}{5} \div \frac{3}{4}$

3.3. $\frac{3}{8} \div 2$

3.4. $\frac{4}{9} \div \frac{8}{3}$

3.5. $5 \div \frac{1}{4}$

3.6. $\frac{5}{7} \div \frac{10}{21}$

3.7. $-\frac{1}{2} \div \frac{3}{5}$

3.8. $\frac{6}{11} \div 3$

3.9. $\frac{7}{10} \div \left(-\frac{14}{15}\right)$

3.10. $-\frac{8}{9} \div \frac{2}{3}$

3.11. $\frac{11}{12} \div \frac{22}{9}$

3.12. $\left(-\frac{4}{7}\right) \div (-8)$

Q4

Calculate the following by first converting the mixed numbers into improper fractions. Write your answer in its simplest form.

4.1. $1\frac{1}{3} + 2\frac{1}{6}$

4.2. $3\frac{1}{2} - 1\frac{1}{4}$

4.3. $1\frac{1}{5} \cdot 2\frac{1}{2}$

4.4. $4\frac{1}{2} \div 1\frac{1}{8}$

4.5. $2\frac{3}{4} + 1\frac{1}{3}$

4.6. $5\frac{1}{8} - 2\frac{3}{4}$

4.7. $-1\frac{2}{5} \cdot \frac{3}{7}$

4.8. $3\frac{1}{4} \div -1\frac{5}{8}$

4.9. $-2\frac{1}{3} - 1\frac{3}{5}$

4.10. $1\frac{5}{6} \cdot 2\frac{2}{11}$

4.11. $-4\frac{1}{5} \div -2\frac{1}{10}$

4.12. $3\frac{1}{9} + 1\frac{5}{6}$

After attempting the questions above, please click this link to find the answers.

Version history and licensing

v1.0: initial version created 12/25 by Donald Campbell as part of a University of St Andrews VIP project.